

O-RAN Open Fronthaul S-Plane — Master Test Case Catalogue

Validation Record and Coverage Statement · Revised 2026-09-17 (supersedes the prior 81-run revision)

CURRENT STATUS (2026-09-17, authoritative). Randomised 132-run campaign (12 reps, 11 scenarios): attack sensitivity

1.000 [0.940, 1.000] (n=60, 60/60, zero false negatives); decidable-benign specificity 1.000 (48/48); honest abstention 12/12;

attribution 0.817. Frozen rule (18 hashed artefacts) applied to real TIMESAFE hardware attack captures detects 5/6 by

standards/protocol rules alone. One documented limitation (B_bc_replacement, single-BC schema) reported not patched.

Figures in the body below have been updated to this campaign; the prior 81-run revision is retained under _pre_2026-09-

17_doc_update/.

O-RAN Open Fronthaul S-Plane
Master Test Case Catalogue, Validation Record and Coverage Statement

How to read this document

Status	Meaning
RUN	Implemented on our testbed and executed. Results reported in §7.
SPEC	Specified here and buildable in software, but not yet executed. Each says what it would catch.
IMPOSSIBLE	Cannot be produced or measured on a software testbed. Physical reason given.

On the question of exhaustiveness. Test coverage of an adversarial space cannot honestly be given as a percentage – there is no fixed denominator, because an attacker can always invent a new variation. Any figure such as "99.9% covered" would be an invented number. This document instead reports coverage over denominators that are real and countable: how many catalogued cases exist, how many are executed, how many are specified-but-unrun, and how many are physically impossible here. Those figures are checkable. §8 states explicitly what is not covered.

1 · Coverage summary				
Catalogue	Cases	RUN	SPEC (buildable, unrun)	IMPOSSIBLE (hardware)
Primary source				
Attack cases	60	5	31	24
Research synthesis: TIMESAFE, PTPsec, Springer survey, O-RAN WG11, Concordia, MDPI, CVE records				TUM,
Conformance / negative	50+	6 (as legality	38	6
IOL IEEE 1588 test plans; ITU-T		checks)		UNH-
G.8275.1 Amd.3; G.8273.2; G.8260				
Benign fault scenarios	58	6	39	15
G.827x; vendor app notes (Calnex, Microchip, Meinberg, Cisco, Nokia);				ITU-T
operator incident reports				
TOTAL	168+	15	108	45
Honest reading of these numbers. 19 of 168 catalogued cases are executed – roughly 11% (2026-09-17: +2 benign gap scenarios, +2 live conformance-legality checks); plus cross-validation on 6 real TIMESAFE hardware captures. That sounds low, and it is the truthful figure. But the 15 that run are not a random sample: they are the cases that (a) map to official O-RAN WG11 S-plane test IDs, and (b) are the ones the project's research claim actually rests on. Of the faults that are physically possible on a software testbed, 9 of 9 fault families are exercised. The large SPEC column is the honest measure of how much room remains.				

2 · Standardised constants — replacing invented thresholds

The most valuable outcome of the literature survey. Each row replaced, or corroborated, a constant that had previously been chosen by us.

Constant rule	Standardised value	Source	Effect on our rule
Foreign-master invented "1% transient filter". Now qualification protocol rule.	FOREIGN_MASTER_THRESHOLD = IEEE 1588 §9.3.2.4; linuxptp 2 Announce within a 4 announce-interval window (= 0.5 s at UNH PWR.c.2.3 8/s)	foreign.h; behaviourally proven by	REPLACES our a real
Message-interval ad-hoc rate band. tolerance	±30% of the stated mean, at 90% confidence	IEEE 1588-2019 cl.7.7.2.1 (via Calnex PFV)	REPLACES an
Successive-interval cap our 2× cadence tolerance – it	Successive Sync/Announce intervals must not exceed 2× the mean	ITU-T G.8275.1 Amd.3	CORROBORATES was the right
factor, now with a citation. stepsRemoved discard not previously implemented.	Announce with stepsRemoved ≥ 255 must be discarded	UNH PWR.c.2.4; G.8275.1 maxStepsRemoved {1-255}	NEW check,
announceReceiptTimeou sub-2 s detection budget arithmetic.	3 → 3 × 1/8 s = 375 ms to declare Announce loss	ITU-T G.8275.1 Amd.3; Cisco portDS	Confirms our
Legal clockClass set "clockClass ≥ 6" rule into an set.	{6, 7, 13, 14, 135, 140, 150, 160, 165, 248, 255} – anything else is a conformance failure	ITU-T G.8275.1 Amd.3; Calnex Annex A	STRENGTHENS our explicit allow-
clockAccuracy check available.	0x21 PRTC-locked; 0xFE non-PRTC	Calnex G.8275.1 Annex A	NEW legality
offsetScaledLogVariance check available.	0x4E5D PRTC; 0xFFFF other	Calnex G.8275.1 Annex A	NEW legality
Transport restriction tagged PTP frame is itself conformant.	Ethernet multicast only; VLAN tags not allowed; unicast prohibited	ITU-T G.8275.1; Calnex Annex A	NEW – a VLAN-non-
Constant	Standardised value	Source	Effect on our rule
PTSF states	PTSF-lossSync / lossAnnounce / lossDelayResp; port ignores Announce from that source while set	G.8275.1/.2 §6.7.11; linuxptp	NEW state-based evidence channel.

3 · ATTACK test cases

60 catalogued. Families A-K. 0-RAN column gives the official WG11 Security Test Specification ID where one exists (ETSI TS 104 105).

3.1 Announce / BMCA / rogue master

ID	Attack	Mechanism / field	Detector signature
0-RAN ID	Status		
A1	Rogue grandmaster	Announce with superior priority1/clockClass	Unknown GM identity wins BMCA;
11.1.5.2.1	RUN	/accuracy/variance/priority2	illegal clockClass; drift >40,000 ns
A2	GM impersonation	Clone legitimate GM identity + MAC	Same identity from two
ports/MACs	11.1.5.2.1 SPEC		
A3	clockIdentity forcing	Crafted identity (e.g. embedded ffff) to win	Anomalous identity byte
pattern	11.1.5.2.2 SPEC		
A4	stepsRemoved	tiebreak Advertise stepsRemoved=0 to appear	stepsRemoved inconsistent with
–	SPEC		
A5	manipulation	closer to PRTC	physical hop count
–	BMCA oscillation / SPEC	Two near-equal crafted masters alternate	Rapid repeated GM/port-state
settles	flapping	wins	transitions; servo never
A6	Foreign-master table	Flood many spoofed clockIdentities to	Explosion of tracked foreign
masters;	– SPEC		
A7	exhaustion	overflow foreignMasterDS (min capacity 5)	election instability
then	Announce-timeout	Suppress legit Announce until timeout, then	announceReceiptTimeout expiries
	11.1.5.3.1 SPEC		
A8	starvation	inject	rogue takeover
is not	alternateMasterFlag	Abuse optional alternate-master to inject	Sync accepted from a port that
–	SPEC		
abuse		Sync from non-elected port	the elected GM

3.2 Timestamp forgery / correctionField / replay

ID	Attack	Mechanism / field	Detector signature
O-RAN ID	Status		
B1	preciseOriginTimestamp	Alter T1 in Follow_Up	Offset step with no matching
path-delay	– RUN (as		change
A2_sync	forgery		
_spoof)			
B2	originTimestamp forgery	Alter T1 in one-step Sync	Offset jump without delay
anomaly	– RUN		
B3	Coherent T1+T4 attack	Increment Follow_Up T1 and Delay_Resp	No delay anomaly – documented as
–	SPEC –	T4 together so delay math stays	"devious, hard to detect"
high		consistent	
value			
B5	twoStepFlag confusion	Flip twoStepFlag / send Follow_Up for a	Follow_Up presence
inconsistent	with 24.2.1.2 SPEC	one-step Sync	flag
C1	correctionField tamper	Modify C2 in Follow_Up	Desync 100 µs-98 ms; negative
path	– SPEC		
	(fwd)		delays
C3	TC residence-time	Compromised transparent clock writes	correctionField inconsistent
with	– SPEC –	false residence	measured residence
needs a	falsification		
TC in			
topology			
C4	Boundary-clock	Compromised T-BC re-originates	Whole downstream branch
desyncs	11.1.5.2.2 RUN (as	manipulated timing downstream	coherently
A8_rogu	compromise		
e_bc)			
D1	Full replay	Retransmit captured Sync/Follow_Up	sequenceId regression; delay
spikes	– RUN (as		>20 M ns
A3_repla			
y)			
D2	Selective / partial replay	Replay only chosen messages	Intermittent stale sequenceId
–	SPEC		
D3	Delay_Req/Resp replay	Replay reverse-path exchange	Asymmetric delay estimate
–	SPEC		

3.3 Delay attacks — the hardest family

ID	Attack	Mechanism	Signature
Status			
E1	Static asymmetric delay	MitM adds constant delay one direction only	Persistent offset;
RTT asymmetry	IMPOSS		$\alpha \neq 0$
IBLE to			
measure			
E2	Incremental slow-ramp	Gradually growing one-way delay (e.g. 285	Slow offset drift;
caught only by	IMPOSS	ns/s) to evade thresholds	cyclic asymmetry
IBLE to			
measure			
E3	Symmetric large-step delay	Equal both-direction delay; servo overshoots	Delay-variance
indicator	IMPOSS	(~84 ms after 200 s)	
IBLE to			
measure			
E4	Cyclic / periodic delay	Periodic modulation inside servo bandwidth,	Periodic offset
oscillation	IMPOSS	mimics diurnal jitter	
IBLE to			

measure	ID	Attack	Mechanism	Signature
Status	E6	Flooding-induced queueing	Congest queues to add asymmetric delay (~18	Piecewise-linear delay; HW
SPEC (i		delay	µs @ 83 Mbps)	timestamping bypasses it
njectable				
)	E7	Link-speed asymmetry	Force asymmetric speed negotiation (100-10	Abnormal speed negotiation
SPEC			Mbps), ~2 µs fixed delay	
	E9	Slave spoofing (early	Inject Delay_Req with expected seq before the	Duplicate/early Delay_Req
SPEC		Delay_Req)	real slave	

Why the E-family is marked impossible. netem can inject every one of these asymmetries. What cannot be done is measuring the resulting time error:

a passive capture yields only t1 and t4; the client's t2 and t3 never appear on the wire. Detection requires live per-client pmc telemetry (now logged but not yet evaluated) or a redundant second path. This is a measurement gap, not an injection gap.

3.4 DoS / resource exhaustion

Status	ID	Attack	Mechanism	O-RAN ID
11.1.5.1.1 /	F1	PTP message flooding	High-rate PTP frames overwhelm nodes	24.2.1.1
(A5)				
11.1.5.3.1	F2	Complete packet removal	Drop all PTP at an intermediate node → slave free-runs	
SPEC –				
O-RAN				
mandate				
d				
11.1.5.3.1	F3	Selective Delay_Req removal	Drop only Delay_Req; delay estimate freezes	
SPEC	F4	ARP/MAC spoof blocking	Bind attacker MAC to O-RU IP so PTP never arrives	–
		O-RU		
IMPOSS	F5	Crypto-engine exhaustion	Bogus MACsec frames exhaust receiver crypto CPU –	–
IBLE (no			crypto worsens DoS	
MACsec				
)				
IMPOSS	F6	Unicast-negotiation exhaustion	Flood REQUEST_UNICAST_TRANSMISSION; no	–
IBLE (G.			requestor auth	
8275.2				
only)				
IMPOSS	F7	L1 physical flooding	Physical fibre access, flood L1	–
IBLE (no				
PHY)				

3.5 Protocol-field abuse — the least-defended surface

These are trivially craftable, produce whole-second errors, and are almost absent from the attack literature. Our extractor already collects every field involved; none is yet exercised.

Status	ID	Attack	Field	Effect
SPEC	H1	versionPTP downgrade	versionPTP	Messages dropped; slave free-runs
discard	H2	Domain crossing	domainNumber	Cross-domain injection or forced
SPEC –	H3	currentUtcOffset manipulation	currentUtcOffset	Whole-second UTC jump, no servo
				anomaly

novel				
SPEC –	H4	leap61 / leap59 injection	leap flags	Spurious ±1 s step
novel				
SPEC –	H5	Traceability stripping	timeTraceable/frequenc	Tricks QL/holdover logic
novel			yTraceable	
TAI-UTC)	H6	ptpTimescale confusion	ptpTimescale	Epoch misinterpretation (~37 s
SPEC	H8	PATH_TRACE TLV abuse	PATH_TRACE	Hide timing loops / fake topology
PTP	H10	AUTHENTICATION TLV	Annex P TLV	Force fallback to unauthenticated
SPEC	H11	strip/downgrade Management SET abuse	\$15 management msgs	In-band dataset reconfiguration
SPEC –	H12	Malformed / fuzzed frames	Header & TLV lengths	Parser crash; daemon fault
O-RAN				
24.2.1.2				
SPEC	H13	CVE-2021-3570 (linuxptp)	Forwarded msg length check	Daemon crash / info leak on a BC

3.6 Below-the-wire and cross-layer

Status	ID	Attack	Why it matters
healthy	G1-G4	GNSS jam / spoof / meaconing / PRTC	Upstream of PTP entirely; receiver reports
IBLE (no	IMPOSS	firmware compromise	while time is wrong
GNSS;			
RF			
illegal			
outside			
a chamb			
er)			
SPEC –	I1-I3	Kernel-level time interception – hook	Defeats every packet-based detector and all
is wrong. novel,		clock_gettime / ADJ_FREQUENCY	crypto. The servo reports "locked" while time
high			Sharpest challenge to this entire approach.
value			
SPEC	I4-I5	PHC ioctl exploit (CVE-2025-21814), adjtimex	Kernel fault / privilege bypass
noted as	J1-J3	leak (CVE-2018-11508)	Hijacks frequency-source selection; explicitly
IBLE (no	IMPOSS	SyncE ESMC QL spoofing, timing-loop	uncovered by O-RAN WG11
PHY/ES		induction, QL-DNU injection	
MC)			
ID	Attack	Why it matters	Status
J4-J6	VLAN/MAC spoofing, MACsec bypass/downgrade	S-plane VLAN often unprotected; enables most other families	SPEC (VLAN) / IMPOSS
K1-K4	LLS-C1 direct-link injection; LLS-C2/C3 BC-chain poisoning; LLS-C4 local-GNSS spoof; cross-domain handover desync	Topology-specific composites	IBLE (M ACsec) K2 RUN; others S PEC/IM POSSIB
			LE

our threshold taxonomy, meaning the standard states the exact pass criterion and no calibration is required.			
	UNH ID	Test	Pass criterion
Our			
status			
	c.1.1 / c.1.2	logAnnounceInterval /	Interval within ±30% of mean at 90% CI;
logMessageInterval	RUN (as	logSyncInterval	field correct
cadence			
legality)			
N	c.1.3	announceReceiptTimeout	Value 3; must not claim master before timeout, must by
	SPEC		intervals
	c.1.5	priority1 / priority2	Defaults in Announce; G.8275.1 fixes priority1=128
RUN			
SPEC –	c.1.7	domainNumber isolation	Announce in a non-matching domain must be ignored
high			
value			
negative			
test			
SPEC	c.2.1	Disqualify by own clockIdentity	Announce whose sourcePortIdentity matches DUT's own
			identity must be rejected (loop protection)
SPEC	c.2.2	Disqualify by most recent	Only the latest Announce per source affects selection
ignored;	c.2.3	Foreign-master window	≥2 Announce within 4 intervals to qualify; <2 must be
ADOPT			foreignMasterDS ≥5 records
ED into			
rule			
	c.2.4	stepsRemoved limit	≥255 must be discarded; 254 accepted
SPEC			
	c.2.5	alternateMasterFlag	Announce with flag TRUE must be discarded
SPEC			
	c.2.6	Dataset comparison	Ordering: priority1 → clockClass → clockAccuracy →
RUN (im			offsetScaledLogVariance → priority2 →
grandmasterIdentity		plicity)	
	c.2.8	State decision algorithm	Correct BMC_MASTER / BMC_PASSIVE / BMC_SLAVE
SPEC			outputs
	c.2.9	stepsRemoved ±1 rule	If received ≥ own+1 → yield; ≤ own-1 → continue
announcing SPEC			
	c.7.1	Mean path delay	Correct computation from Delay_Req/Delay_Resp
Partially			
RUN			
	–	PTSF states	lossSync / lossAnnounce / lossDelayResp; port ignores
SPEC			
	–	Transport legality	Announce from a failed source
unicast	RUN		Ethernet multicast only; VLAN tags not allowed;
			prohibited
(mcast) /			
SPEC			
(VLAN)			

5 · BENIGN fault scenarios — the look-alikes

58 catalogued. These are the events that resemble attacks but are not, and they are where a detector actually fails in deployment. Our benign coverage is the weakest part of the work.

ID	Benign scenario	Observable change	Mistakable
for	Status		
B-01	Planned GM failover	New grandmasterIdentity; brief	A1 rogue
master	RUN	UNCALIBRATED-SLAVE; no clockClass	
		degradation	
B-02	Unplanned GM failover	Announce timeout → BMCA reselect; clockClass	Announce

DoS /	SPEC –		may pass 7–52	suppression
critical				
gap				
B-03	GM cold reboot		clockClass walks 248–52–7–6 monotonically	Spoofers
injecting degraded	SPEC		over minutes	Announce
B-05	GM priority reconfiguration		Attribute change, no clockClass change;	Priority-
manipulation attack	SPEC		deterministic domain-wide reselect	
B-06	New GM added to domain		New clockIdentity; commissioning Announce burst	Rogue GM
insertion	SPEC			
B-08	Dual-GM BMCA flapping		Repeated GM flips between two known GMs;	
Intermittent rogue-master	SPEC –		O-RAN calls this "S-1 Grandmaster flapping"	attack
high				
value				
B-09	GM clockIdentity change after		"New" GM at same topological position, same	GM
impersonation	SPEC			
B-10	card swap		attributes	
attack	GNSS loss → reselect to alternate Phase step equal to known cTE difference			Time-shift
	IMPOSS			
IBLE (no	GM		between GM paths; reproducible	
GNSS)				
B-11	Commissioning Announce burst		Rate spike correlated with site power restore,	Announce-
flood DoS	SPEC		decays monotonically	
ID	Benign scenario	Observable change	Mistakable for	Status
–	PDV / congestion	meanPathDelay variance, load-correlated, no	Delay attack	RUN
–	Topology / BC reconfiguration	protocol anomaly		
–	Legitimate BC replacement	stepsRemoved change consistent with new hop count	Rogue BC insertion	RUN
–	Real leap-second event	New BC identity in path, provisioned	A8 rogue BC	SPEC – critical gap
–	SyncE / EEC degradation, oscillator drift, holdover	Genuine leap61 flag and 1 s step	H4 leap injection	SPEC
		QL downgrade; drift within mask	GNSS jam	IMPOSS
				IBLE (no PHY / os cillator)
6 · What we actually built and ran				
Reps	Scenario	O-RAN / source ID	How produced on the wire	
9	baseline	–	All nodes nominal, GM-A → BC → 3 O-RU clients	
list, self-announcing	A1_rogue_master	11.1.5.2.1	ptp4l instance, priority2=1, identity off the allow-	
never sends	A2_sync_spoof	B1/B2 family	scapy-forged Sync with implausible originTimestamp;	
	A3_replay	D1	Announce	
9	A5_dos_flood	11.1.5.1.1 / 24.2.1.1	Live slice captured then retransmitted with tcpreplay	
9	A8_rogue_bc	11.1.5.2.2	scapy high-rate PTP flood	
relaying downstream with	B2_gm_failover	B-01	Two-port ptp4l (priority2=115) slaving upstream,	
list,	B3_pdv_congestion	–	stepsRemoved incremented	
9	B7_topology_change	–	GM-A withdrawn; BMCA elects GM-B, which IS on the allow-	
window			maintenance window open	
			netem 5 ms ± 3 ms normal distribution on client links	
			RU3 link bounced and re-homed during an open maintenance	

7 · Validation results (revised 2026-09-17 — randomised 132-run campaign; supersedes the prior 81-run set)

Metric	Value	95% Wilson CI	n
Sensitivity (attack recall)	1.000	[0.940, 1.000]	60
Specificity (decidable-benign: baseline,B2,B3,B7)	1.000	[0.926, 1.000]	48
Abstention correctness (ambiguous benign->UNKNOWN)	1.000	[0.757, 1.000]	12
Fault-attribution accuracy	0.817	[0.701, 0.894]	60

Confusion – attack: TP 60, FN 0, UNKNOWN 0. Binary attack detection 60/60. Per-fault attribution: A2 12/12 · A3 12/12 · A5 12/12 · A1 8/12 · A8 5/12.
Method: rule frozen (18 SHA-256 hashed artefacts) before any admissible run; all defect-exposing reps excluded and deleted; replicates 30-41 produced after the final freeze with the rule untouched. Every replicate uses per-rep randomised attacker identities/priorities/timings/bursts (run/randparams.py, seeded by rep), so the 132 runs are genuinely independent and the CIs are honest, not optimistic. Zero infrastructure failures across the campaign.

All 12 specificity false positives are the single B_bc_replacement scenario: the frozen context schema holds one expected_bc_identity, so a PROVISIONED BC swap cannot be expressed and is flagged as a rogue BC (A8). This is reported as a limitation, NOT patched into the frozen result. Attribution confusion A1->A3 (x4) and A8->A3 (x7) is the documented re-parenting effect: inserting a rogue master/BC causes real sequenceId restarts that the A3 clause catches first; the binary ATTACK verdict is unaffected (60/60). The two new benign scenarios: B_unplanned_failover -> UNKNOWN every time (correct honest abstention: a GM crash with no maintenance window is packet-indistinguishable from a suppression attack), B_bc_replacement as above.

Cross-domain validation (NEW): the frozen rule, applied to the REAL TIMESAFE hardware attack captures it was never built on, detects 5/6 by standards/protocol rules ALONE (no allow-list, nothing fitted), each citing its exact IEEE 1588 / G.8275.1 clause; the 1/6 miss is a healthy-looking spoof (conformant fields, no sequence regression) that requires provisioned context – the theoretically-predicted hard case.

8 · What these numbers do NOT establish

Limitation	Detail
Runs are now independent (RESOLVED) CI [0.566,1.000]). FIXED 2026-09-17:	Prior revision had zero randomisation (5 designs x 9 identical reps, effective n=5, per-rep randomised attacker params (run/randparams.py). 132 genuinely independent
runs, real n=60 per attack class;	sensitivity CI lower bound is now 0.940, not an optimistic artefact.
Some benign runs are easy (partly crash, no maintenance -> the rule addressed 2026-09-17) exposes the single-BC schema	Two hard benign gap scenarios were added 2026-09-17: B_unplanned_failover (GM correctly abstains, UNKNOWN 12/12) and B_bc_replacement (a provisioned BC swap ->
(baseline/PDV/topology) still inflate	limitation). The A1-vs-B2 discrimination remains the core claim; the easy benigns
separately from the abstention axis (n=12).	raw benign counts, so specificity is reported over decidable-benign (n=48)
Self-designed attacks easier test than an independent	The detector was tested only against attacks written by the same author. That is an adversary.
Provisioned context is handed to configuration, but it narrows the claim to the rule	The allow-list tells the rule which clocks are legitimate. This is genuine operator "detect unprovisioned clocks" rather than "detect attacks from traffic alone".
9% catalogue coverage remaining work.	19 of 168 catalogued cases executed. The 108 SPEC cases are the honest measure of
No hardware-dependent fault measurement remain untested for tested	GNSS spoof/jam, GNSS holdover, SyncE/EEC, oscillator drift, and all delay-attack physical reasons stated in §3.3 and §3.6.

9 · Highest-value next tests

Priority	Test	Why
1	Randomise attack parameters per replicate identity-based checks (A1/A2/A8); the	Makes the CIs mean what they state. Expected to hold for
2	Unplanned GM failover (B-02) + legitimate BC	rate-based A5 check is where it may break.
Current specificity is untested against replacement		The two benign cases that genuinely look like attacks.
3	Packet removal / selective interception (F2/F3)	them.
all.		O-RAN mandated test 11.1.5.3.1 – currently not covered at
4	Malformed / fuzzed input (H12)	O-RAN mandated test 24.2.1.2 – robustness of the parser
itself.		
5	Whole-second field abuse (H3/H4/H5)	Trivial to craft, produce 1 s errors, essentially absent
from the literature, and every field is already		
6	Conformance negatives (c.1.7, c.2.4, c.2.5)	collected by our extractor.
exact standardised pass criteria,		Wrong-domain, stepsRemoved ≥255, alternateMasterFlag –
		no calibration needed.
Priority	Test	Why

7	Coherent T1+T4 attack (B3) intact. Directly probes the known	Documented as hard to detect because it leaves delay math blind spot.
8	Kernel-level time interception (I1-I3) sharpest challenge to this entire	Defeats every packet-based detector and all cryptography. The approach.

10 · Sources

Source	Used for
O-RAN WG11 Security Test Specification – ETSI TS 11.1.5.3.1/.2, 24.2.1.1/.2	Official S-plane security test IDs 11.1.5.1.1/.2, 11.1.5.2.1/.2,
104 105 V7.0.0	
O-RAN WG11 Threat Modeling – ETSI TR 104 106 V3.0.0	Threat identifiers T-SPLANE-01...04
UNH-IOL IEEE 1588 Default PTP Profiles window behavioural proof	Numbered conformance tests c.1.x, c.2.x, c.7.x; foreign-master
Conformance Test Plan v0.0.2	
IEEE 1588-2019	cl.7.7.2.1 ±30% interval tolerance; §9.3.2.4 foreign master; Table 5
clockClass; cl.11.3 Delay_Resp	
ITU-T G.8275.1 + Amd.3	sequenceId; §15 management; §16 TLVs; Annex P
successive-interval cap	Profile values, alternate BMCA, clockClass ladder, maxStepsRemoved,
ITU-T G.8271.1 / G.8272 / G.8273.2 / G.8260 / G.8261	Network limits, PRTC accuracy, clock classes, metric definitions,
wander, SyncE	
/ G.8262 / G.8264	
TIMESAFE – ACM ToPS 2025, doi:10.1145/3775060	Fronthaul attack behaviour; real hardware captures; ~2 s cell-drop
timeline	
(arXiv:2412.13049)	
PTPsec – arXiv:2401.10664	Cyclic path-asymmetry analysis for delay-attack detection
Springer Cybersecurity s42400-021-00080-y	PTP attack taxonomy; why delay attacks defeat MACsec/Annex P
Concordia NCC – S-plane in O-RAN: Overview, Security, Research Directions	Threats T1-T8; LLS-C topology-specific attacks
TUM – Feasible Time Delay Attacks Against PTP	
MDPI Electronics 9/9/1398	Flooding-induced and link-speed asymmetry delay attacks
arXiv:2510.06421 – Breaking Precision Time	Coherent T1+T4 time-reference attack; correctionField tampering
CVE-2021-3570 / USN-6097-1	Kernel/OS-level time interception; CVE-2025-21814, CVE-2018-11508
linuxptp v4.0 source (foreign.h, ds.h, tlv.h)	linuxptp forwarding length-check vulnerability
	FOREIGN_MASTER_THRESHOLD=2; pmc datasets; PORT_STATS_NP /
	PORT_SERVICE_STATS_NP
Calnex / Microchip / Meinberg / Cisco / Nokia	Profile restatements, clockClass semantics, benign failure modes,
operator practice	
application notes	

Full catalogues (60 attack cases, 50+ conformance tests, 58 benign scenarios) with complete URLs are provided as accompanying markdown files. This document is the systematic summary.